11.4.2 Particle Interaction

In particle physics simulations, scientists need to model systems where a high number of atoms, molecules or sub-atomic particles interact in a closed environment. For instance, you could simulate the evolution of a gas when temperature change, or a laser beam hits the gas, etc…

Figure 11.10 shows a simplification of what such a simulation could like. Considering that, at an average 25 °C room temperature, there are approximately 10^{22} molecules in a square meter of air, you can imagine that even with small boxes and rarefied gases, your simulation should handle billions of billions of items, for each step of the simulation.

These particles simulations are all about computing the interaction between particles: but with such numbers, checking how each particle interacts with any other particle is simply not feasible: there would be ~10^{40} pairs to check, and that number is just too big for any traditional computer[18].

Listing 11.10 Particle Interaction with Range Search

```
function simulation(numParticles,
simulationComplete)                          #1
  particles ←
initParticles(numParticles)                                  #2
  while (not
simulationComplete)
   #3
    tree ← S-tree(particles, 3)
                                    #4
```